Joel Rogers

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PROFESSIONAL SUMMARY

Penultimate year **Mechatronics Engineering student** at the University of Canterbury, I have a passion for aerospace and industry experience in **PCB Design, PCB assembly, firmware programming (C++), hardware integration**, **automation (PLC)** and **working with aerospace technicians/engineers**. During my time at Swarm Intelligence, I was on site at the Christchurch Engine Centre (Pratt & Whitney) numerous times working with their technicians and engineers commissioning one of the industrial automation projects I programmed (PLC). I have developed two flying wing UAVs. I designed and assembled a flight controller **PCB** with IMU, GPS and RF modules for autonomous flight.

EXPERIENCE

Professional

Electronics/Mechatronics Engineering, Swarm Intelligence
Christchurch, Canterbury, March 2026-August 2026

- **Soldered** over 30 PCBs, both surface mount and through hole.
- **Programmed** and **commissioned** automated manufacturing systems in high performance engineering environments (including projects with the **Christchurch Engine Centre** and **Fulton Hogan**).
- Developed an ESP32 based multiprotocol **wireless communicating** industrial monitor. This incorporated **GPS tracking**, **firmware programming** (C++) and **hardware integration.** Communicates over **4G**, **WiFi** and **Ethernet**.
- **Communicated** with our industry partners to raise awareness of our new product, creating briefs and participating in client meetings.
- **PCB design** (**KiCAD**) for microcontroller boards in real world mobile systems.
- **Programmed** a **database**, **server**, **data parsing** (JavaScript) and **web UI** (HTML) for the wireless industrial monitoring system.

Electronics Engineering Intern, Swarm Intelligence
Christchurch, Canterbury, October 2025-March 2026

- **Validated** all assembled PCBs to verify correct operation and when necessary, **debugging** faults prior to delivery.
- Designed multiple **HMIs** for the technicians at the Engine Centre and for the clients of the ESP32 industrial monitor.
- Completed on site **hardware installation and commissioning** teaching me the safety requirements and **debugging** process when dealing with **high voltage power electronics, pumps** and **pressure vessels.**

Technical Projects

UAVs (Personal)

- Manufactured a custom **fibre glass composite** fuselage core to provide a lightweight, high-strength structural section.
- Designed a **flight controller** (**KiCAD**) with integrated **GPS** and **IMU** modules for **fly by wire** assistance and **autonomous** flight.
- Designed through iteration, modelled (**SolidWorks**), manufactured and tested two flying wing UAV platforms, the prototype and the V2.
- Integrated **RF** communication for command and control.
- **Programmed** the flight controller to take inputs from the **RF**, **GPS** and **IMU** modules to control the plane accordingly.

Line Following Robot (University)

- **Broke the record** for the annual University of Canterbury line following robot competition, with our fastest time being **5.81s**.
- Designed (**KiCAD**) and assembled a **microcontroller PCB** and smaller photoelectric sensor PCBs.
- Programmed the robot in **Arduino IDE** and integrated an ESP8266 for **wireless communication** and PID value changes over **WIFI**.

RoboCup (University)

- Integrating **mechanical**, **electrical** and **embedded** systems to develop an **autonomous** robot that navigates dynamic environments, identifies and collects target objects, to compete in the University of Canterbury RoboCup competition.
- Using **first principles** engineering to achieve a relatively simple robot with a very reliable weight pick up mechanism.

Solar Car Power Supply PCB (University)

- **Designed** a buck converter, PWM generator and comparator **PCB** (KiCAD) to supply power from a solar panel to two DC motors. (ENCE372 **Power Electronics**)
- Calculated the values of capacitors, resistors and an inductor required to generate the correct voltage step down. Assembled the **PCB** (**soldering**).

STM32 Step Counter (University)

- Programmed (**STM32CubeIDE**) an STM32 walking step counting system using **interrupt** driven software.
- Steps were detected through reading an **IMU** and implementing threshold based counting.
- Utilized **GIT** for version control.

SKILLS

Programming:

- Python
- C, C++, C#, Arduino
- HTML
- JavaScript
- GIT

Embedded & Electronics:

- KiCAD
- Soldering and electronic assembly
- FactoryTalkView
- Sensor integration
- AutoCAD Electrical
- CX Programmer
- STM32CubeIDE
- Arduino IDE

Communication protocols and Networking:

- CAN bus
- Ethernet
- SMS
- GNSS tracking
- MQTT
- Modbus
- RF

Mechanical:

- SolidWorks
- 3D printing
- Fusion 360
- Lathe
- Mill

Soft Skills:

- **Communication:** meetings with clients and coworkers at Swarm Intelligence. The ability to communicate technical ideas to non-technical people.

Further Experience

- **Workplace First Aider** Certified (Valid until February 2028)

Professional:

- **Tiler**, Syrus Imperial Tiles (Dec 2023 - Nov 2025)
- **Bartender**, The Commodore Hotel (November 2022 - February 2025)
- **Hockey Coach**, St Andrews College (April 2023 - August 2025)
- **Representative Hockey Coach,** Canterbury Hockey (August 2025 - September 2025)

EDUCATION

Bachelor of Engineering (Honours) in Mechatronics
Christchurch, University of Canterbury,
Expected graduation: 2027

Website: https://j0llyr0ger.github.io/Joel-Rogers-Portfolio/

References:
- **Craig Blaikie**, CTO - Swarm Intelligence, 021386817
- **Harry Davis**, Electrical Engineer - Swarm Intelligence, 021386812
- **Jonny Bovaird**, Key Accounts Manager - Christchurch Engine Centre, 0272538926

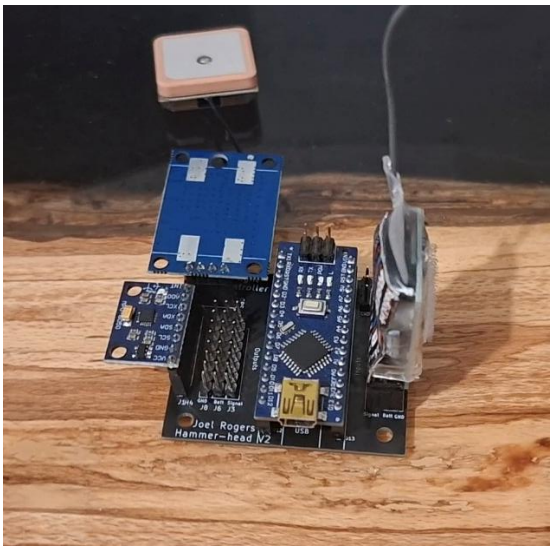
Joel Rogers

Mechatronics Engineering Portfolio

UAV V2



Fabricated an ultra-strong **fibre glass** core section (light grey) for strength, small wall thickness and weight.



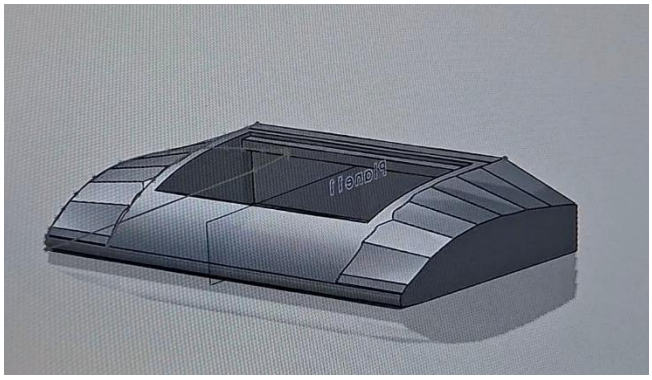
Designed a **flight controller** (**KiCAD**) with integrated GPS, RF receiver and IMU module for autonomous flight and fly by wire pilot assistance.



Hand launch of the UAV V2.



A display of the **fly by wire assistance** where the elevons are attempting to relevel the plane.



Designed, modelled (**SolidWorks**), manufactured and tested a flying wing UAV platform.



Design changes through the learning of the prototype (**design through iteration**).

UAV Prototype



Designed, modelled (**SolidWorks**), manufactured and tested a flying wing UAV platform.



Integrated long range **RF** communication for **command and control.**



Iterating the design through failure and partial destruction.

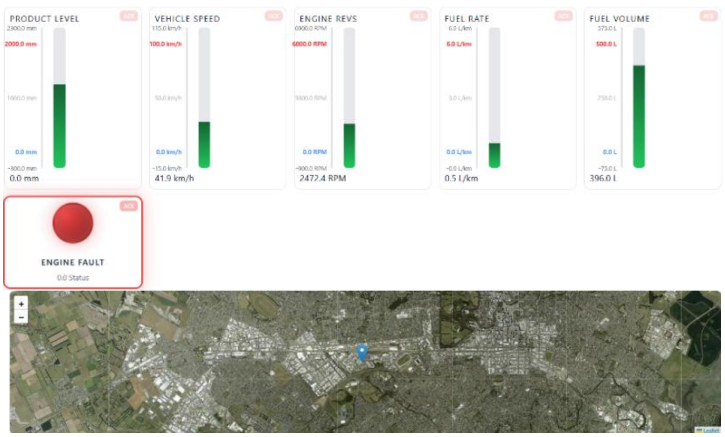


Achieved flight after six failed attempts and subsequent redesigns.

Sentinel Wireless IoT Module

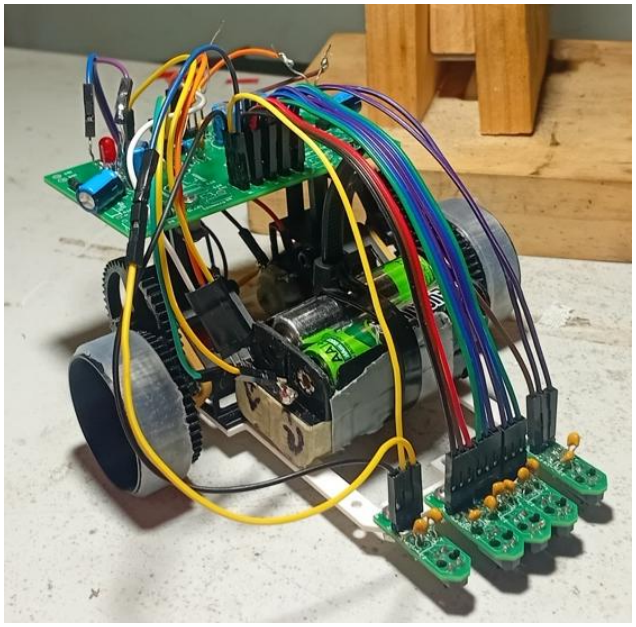


Multiprotocol wireless communicating IoT ESP32 gateway, communicating through **WiFi**, **4G, Ethernet** and obtaining **GPS telemetry**.

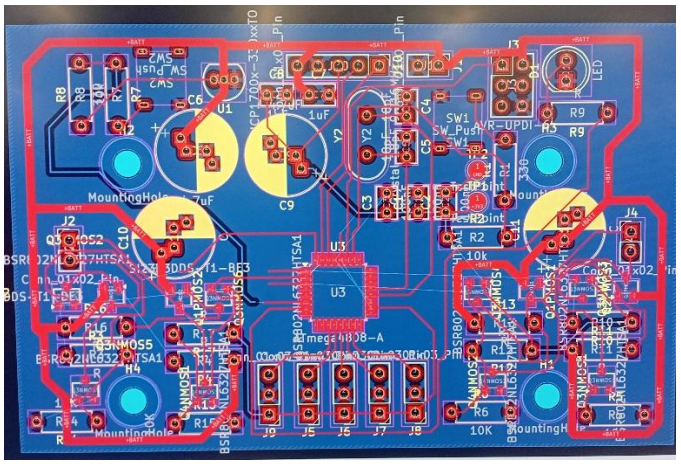


Also included the construction of a **server, database** and custom **web UI**.

Line Following Robot



Broke the record for the annual University of Canterbury line following robot competition, with our fastest time being **5.81s**. Designed, 3D modelled (**Fusion 360/SolidWorks**), **3D printed** and tested numerous gearing systems, motor mounts and chassis.

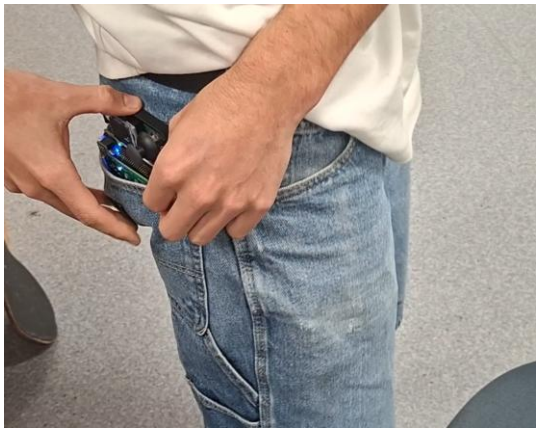


Designed (**KiCAD**) and assembled the main **microcontroller PCB** and smaller sensor PCBs. Programmed the robot in **Arduino IDE** and integrated an ESP8266 for **wireless communication** and PID value changes over **WIFI**.

STM32 Step Counter



Programmed (**STM32CubeIDE**) an STM32 walking step counting system using **interrupt** driven software. Utilized **GIT** for version control.

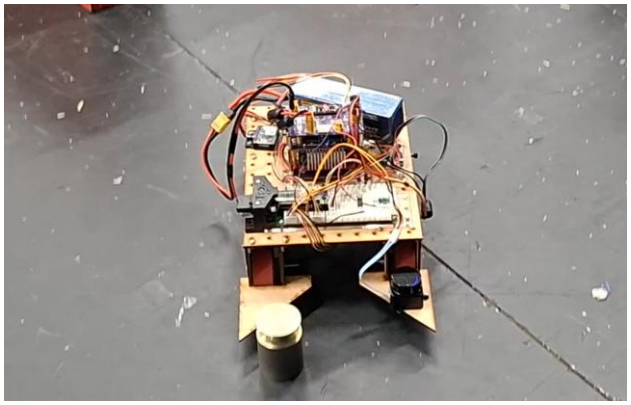


Steps were detected through reading an **IMU** and implementing threshold based counting.

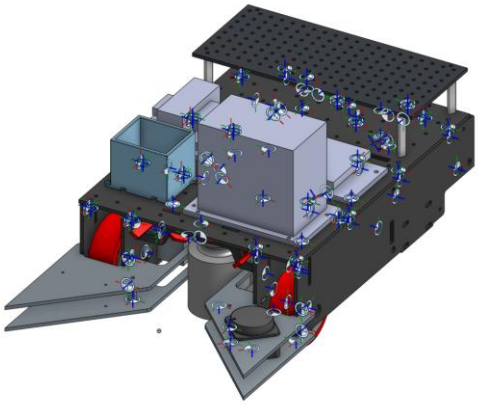
RoboCup



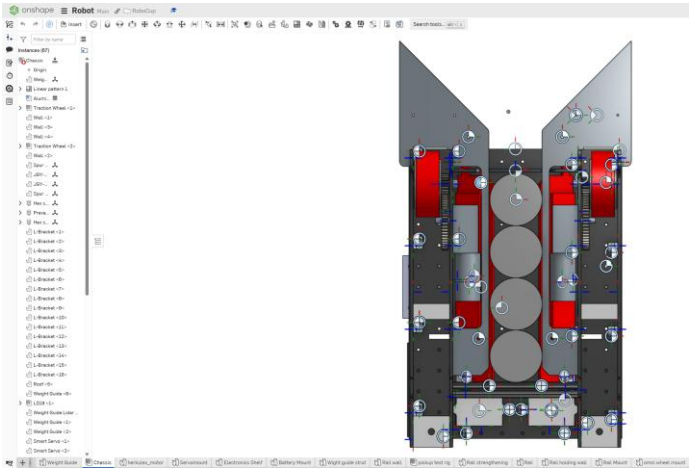
Used **first principles** engineering to design and **iterate** a weight collection system,



Developing a **MDF** chassis using **Laser cutting** and **3d printing** to collect metal weights around an arena.



3d modelled using **OnShape**.



Using **OnShapes** version control for ease of **redesign** and **reverting changes**.